A Coupled Radiative–Microphysical Model of Titan's Organic Haze: Energy Balance, a Fractal-Aggregate Scaling Law, and Their Feedback

The Titan Haze Feedback Project

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Abstract

Titan's organic haze is radiatively and microphysically active: it absorbs sunlight and warms the stratosphere, emits in the thermal infrared, and its vertical and size structure is set by coagulation and sedimentation of fractal aggregates whose rates depend on the local temperature. This makes the haze a genuine feedback element rather than a passive tracer. We outline a four-stage program to isolate this feedback in a one-dimensional column: (i) a multi-stream discrete-ordinates (DISORT/`pydisort`) radiative energy balance for the temperature profile; (ii) an analytic *scaling law* for the aerosol microphysics that maps temperature, monomer radius, and fractal dimension onto the haze number- and size-profile; (iii) iteration of (i) and (ii) to a self-consistent fixed point to diagnose the feedback; and (iv) coupling of photochemical haze production into the loop. Our central methodological result is the scaling law. Adopting a monodisperse, self-similar closure $C(z,N) = n(z)\,\delta(N - \bar{N}(z))$ for the concentration of aggregates of N monomers at altitude z, the population balance reduces from an integro-differential equation in (z, N) to a compact boundary-value problem in z for the number density $n(z)$ and mean size $\bar{N}(z)$. In the sedimentation-dominated limit it collapses further to a single first-order ordinary differential equation, $\mathrm{d}\bar{N}/\mathrm{d}z = -\beta(\bar{N}, z)\,P/[2\,\omega(\bar{N}, z)^2]$, whose asymptotic solutions grow as $\bar{N} \propto (\text{depth})^2$ in the free-molecular upper haze and $\bar{N} \propto (\text{depth})^{1/2}$ in the continuum lower atmosphere. Solving the full eddy-diffusion problem reproduces the haze extinction scale height of Tomasko et al. (2008) (64 vs. 65 km) and a sub-micron characteristic radius consistent with de Trenquelléon et al. (2025). This law is inexpensive enough to embed inside the radiative iteration, closing the feedback loop.

1 Introduction

On Titan the photochemical haze is not a passive optical load. It absorbs solar radiation—heating the stratosphere—and emits in the thermal infrared, where it contributes a non-negligible fraction of the planetary emission and warms the troposphere (Tomasko et al., 2008). At the same time its vertical extent and particle sizes are governed by aerosol microphysics: monomers produced high in the atmosphere coagulate into fractal aggregates that sediment downward (Cabane et al., 1993; Rannou et al., 2004; Burgalat and Rannou, 2017). Crucially, the microphysical rates are temperature dependent—the Brownian coagulation kernel scales as T/η, the gas viscosity $\eta(T)$ and mean free path $\lambda(T, P)$ enter the settling velocity, and the free-molecular kernel carries a $T^{1/2}$ factor. Temperature sets the microphysics, the microphysics sets the opacity, and the opacity sets the temperature: a closed feedback loop.

Most existing models break this loop in one direction. Radiative studies prescribe the haze from observations (Tomasko et al., 2008; Lombardo and Lora, 2023), while fully coupled treatments embed microphysics in a three-dimensional general circulation model (de Batz de Trenquelléon et al., 2025), which is faithful but expensive and difficult to interrogate. Our aim is intermediate: to isolate the one-dimensional radiative↔microphysical feedback with a fast, analytic microphysics so the coupled system can be iterated cheaply to a self-consistent state and the feedback gain measured directly.

2 Model overview

The program proceeds in four stages.

Step 1 — Radiative energy balance. A 1-D plane-parallel Titan column solved with a multi-stream discrete-ordinates method (`pydisort`/DISORT; Stamnes et al., 1988). Shortwave absorption by CH_4 and haze; longwave cooling by gas lines, collision-induced absorption, and haze emission. Iterated to radiative(–convective) equilibrium for $T(z)$ (Sec. 3).

Step 2 — Microphysical scaling law. A closed relation for the steady aggregate population, coagulation + sedimentation → density/size profile, in the form $C(z, N)\,\mathrm{d}z\,\mathrm{d}N = f(z, N, T, d, D)$ (Sec. 4). This is the methodological core of the present note.

Step 3 — Coupled iteration. Map the microphysical moments to layer optical properties, feed DISORT, update $T(z)$, and iterate to a fixed point; diagnose the feedback (Sec. 5).

Step 4 — Photochemistry. Promote the imposed haze production rate to an output of a photochemical module, closing the radiation↔chemistry↔microphysics loop (Sec. 6).

We note that none of the reference studies (Tomasko et al., 2008; Lombardo and Lora, 2023; de Batz de Trenquelléon et al., 2025) use DISORT—they employ doubling–adding, line-by-line, or two-stream solvers—so the multi-stream solver here is an upgrade, validated against the net-flux and heating/cooling tables of Tomasko et al. (2008).

3 Radiative energy balance (Step 1)

The column spans the surface to $\sim 540\,\mathrm{km}$. Following the spectral split of Lombardo and Lora (2023), shortwave (2000–40000 cm^{-1}, $< 5\,\mu\mathrm{m}$) radiative transfer treats solar absorption and multiple scattering by haze (single-scattering albedo ω_0, asymmetry g), while the longwave band (1–2000 cm^{-1}) handles thermal cooling with gas line opacity, N_2–N_2, N_2–CH_4, CH_4–CH_4, and N_2–H_2 collision-induced absorption, and a (near-)purely-absorbing haze. Gas opacities use the correlated-k method with HITRAN line data; methane shortwave coefficients follow the Huygens/DISR measurements supplemented above 1600 nm by laboratory data. The temperature profile is relaxed until shortwave heating balances longwave cooling.

The validation target is the energy budget of Tomasko et al. (2008): a surface net solar flux of $0.574\,\mathrm{W\,m^{-2}}$ (about 11% of the incident flux) rising to $\sim 4.1\,\mathrm{W\,m^{-2}}$ near 400 km; a net heating that exceeds cooling by at most $\sim 0.5\,\mathrm{K}$ per Titan day near 120 km; and a thermal cooling rate peaking near 21 K per Titan day around 340 km. Roughly 80% of the top-of-atmosphere sunlight is absorbed in the column, $\sim 40\%$ below 80 km, the haze raising the planetary thermal emission by $\sim 15\%$ and reducing the tropospheric cooling rate by $\sim 30\%$ below 50 km. Baseline atmospheric and compositional inputs are collected in Table 2.

3.1 Implementation and first results

The solver is built on `pydisort` (a modern `torch`-based DISORT binding) with eight streams: a collimated solar beam in the shortwave and a band-resolved Planck thermal source in the longwave. Per layer the optical properties combine the haze—its extinction from the Step 2 microphysics (cross-section $Q_{\rm ext}\pi r_a^2$), with *spectral* single-scattering albedo $\omega_0(\lambda)$ and asymmetry $g(\lambda)$ taken from the Huygens/DISR prescribed haze (Doose 2016): $\omega_0 \to 0$ in the UV (pure absorber), rising to ~ 0.9 in the near-IR, and a near-pure absorber in the thermal IR—and the gas.

The longwave combines two sources on a common 40-band IR grid (0–2000 cm^{-1}, 50 cm^{-1} bands). First, *real collision-induced absorption*—the dominant thermal-IR continuum on Titan—from the HITRAN CIA coefficients (Karman et al., 2019) for N_2–N_2, N_2–CH_4, CH_4–CH_4, and N_2–H_2, as $\tau(\nu) = \sum_{\rm pairs} k_{AB}(\nu, T)\, n_A n_B\, \Delta z$; the N_2–N_2 roto-translational band makes the far-IR strongly opaque (column $\tau \gtrsim 100$ near 50–100 cm^{-1}), the origin of Titan's pressure-induced greenhouse. Second, *correlated-k gas lines* for the stratospheric coolants CH_4, C_2H_2, C_2H_6, C_2H_4, and HCN (the same IR k-tables and 10-Gauss grid as the shortwave), summed under the perfectly-correlated assumption with each gas's VMR profile. Both gases and CIA, plus a pure-absorber IR haze, are solved per (band, Gauss-point) with a band-integrated Planck source.

The *shortwave* methane opacity is treated with correlated-k, using the same visible CH_4 k-table as the reference Fortran model (42 bands × 10 Gauss points, on a 20 × 10 pressure–temperature grid). Per layer the gas optical depth is $\tau = k(P, T)\, u$ with u the CH_4 column abundance in km-amagat; all 420 (band, Gauss-point) sub-intervals are solved in a single multi-wave DISORT call and recombined by the Gauss weights, with the per-band solar flux integrated from the Houghton spectrum and scaled to Titan's orbit ($\sum$ bands $= 14.8\,{\rm W\,m^{-2}}$). The CH_4 windows let sunlight through to the surface while the bands absorb aloft; CH_4 alone absorbs $\approx 1.4\,{\rm W\,m^{-2}}$, peaking at ~ 14 K per Titan day in the stratosphere.

We relax the *layer-mean* temperature toward radiative–convective equilibrium: each iteration runs both bands, nudges the layer temperatures along the net heating rate (under-relaxed; the layer-mean state is in 1:1 correspondence with the per-layer heating, avoiding the level↔layer aliasing that otherwise seeds a grid-scale zig-zag), then applies a convective adjustment to a critical lapse rate ($\sim 1\,{\rm K\,km^{-1}}$). Cold space is imposed as the top boundary (a thermal emission temperature of 2.7 K, not the atmosphere's top temperature); without this the column spuriously absorbs downwelling longwave and overheats.

The solver reaches a genuine steady state with $\sim 4.0\,{\rm W\,m^{-2}}$ of shortwave absorbed (close to the $\sim 4.5\,{\rm W\,m^{-2}}$ of Tomasko et al., 2008), energy closed to $\sim 0.1\,{\rm W\,m^{-2}}$. Driven by the Step 2 microphysics haze it has the right vertical structure—a stratopause (~ 300 Pa), a cold tropopause near 74 K, a ~ 92 K surface—but a stratosphere ~ 80 K too cold (Fig. 1, red).

To localize this deficit we ran the *same* radiative engine on the observationally-prescribed haze (the optical depths the reference model uses) instead of the microphysics haze. It then reaches a ~ 200 K stratopause, matching the reference (Fig. 2, green vs. black) — so the radiation physics (correlated-k CH_4 shortwave, spectral haze, CIA, correlated-k gas-line longwave, convective adjustment, energy closure) is *correct*, and the entire ~ 80 K cold bias is the microphysics haze: our Step 2 haze places too much optical depth too low and too much in total (the mobility-radius optical cross-section overestimates large- aggregate extinction, giving a column optical depth several times observed). This is a Step 2 microphysics→optics calibration issue (mean-field aggregate optics), not a radiation one. The validated radiation engine, energy closure, and haze→optics→DISORT coupling are in place, which is what Step 3 requires.

3.2 Cross-comparison with a reference radiation model

As an independent check we compare against a mature Fortran radiation model of the TAM lineage (Lombardo and Lora, 2023, the same scheme family as Lombardo & Lora 2023), a full correlated-k treatment (CH_4, C_2H_2, C_2H_6, C_2H_4, HCN plus N_2–N_2/N_2–CH_4/CH_4–CH_4/N_2–H_2 CIA) with prescribed haze, run to radiative–convective equilibrium on a 100-layer column under

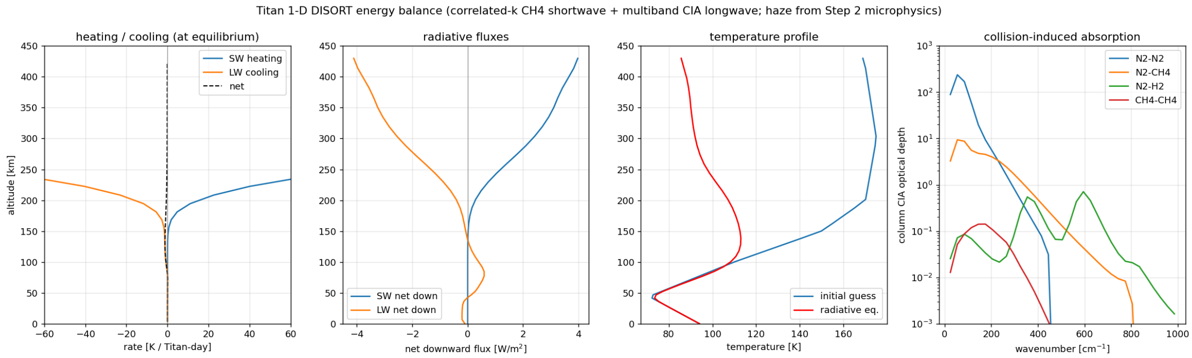

Figure 1: DISORT energy balance on the Titan column with the Step 2 haze and HITRAN collision-induced absorption. From left: shortwave heating, longwave cooling, and net rate at the relaxed state; net downward shortwave and longwave fluxes; initial-guess vs. radiative-equilibrium temperature (surface fixed); and the column CIA optical depth per pair, showing the N_2–N_2 far-IR continuum that dominates the thermal opacity.

diurnally-averaged insolation (the diurnal cycle disabled for faster convergence). Both models are at steady state.

Figure 2 overlays three profiles on a shared pressure coordinate. Driven by the Step 2 microphysics haze (red), our model has the right structure but a $\sim 115\,\mathrm{K}$ stratopause versus the reference's $\sim 195\,\mathrm{K}$. Driven instead by the *prescribed* haze (green)— the observational optical depths the reference uses, in the same radiative engine—our model reaches $\sim 200\,\mathrm{K}$ and tracks the reference (black) closely through the stratosphere and mesosphere. This isolates the discrepancy cleanly: the radiation is right (the engine reproduces the reference given the same haze), and the cold bias of the coupled run is entirely the microphysics haze—its optical depth is too large and deposited too low (the mobility-radius optical cross-section overestimates large-aggregate extinction). The remaining upgrade is therefore in Step 2 (mean-field aggregate optics for the haze cross-section), not in the radiation.

4 Microphysical scaling law (Step 2)

We seek the steady population $C(z, N)$ of aggregates containing N monomers at altitude z (positive upward), built from production, Brownian coagulation, and gravitational sedimentation, with T the temperature, d the monomer radius, and D the fractal dimension. The physical ingredients—kernels, settling velocity, and fractal radius laws—follow Burgalat and Rannou (2017), with parameters from de Batz de Trenquelléon et al. (2025).

4.1 Monodisperse, self-similar closure

We adopt a monodisperse closure: at each altitude the aggregates share a single characteristic monomer count $\bar{N}(z)$,

$$C(z, N) = n(z)\,\delta\big(N - \bar{N}(z)\big)\,, \tag{1}$$

so the unknown fields are the number density $n(z)$ $[\mathrm{m}^{-3}]$ and the mean size $\bar{N}(z)$. The count $\bar{N}$ is treated as a *continuous monomer-volume count*: it measures particle volume in units of one monomer's volume, so $\bar{N} < 1$ denotes a sub-monomer sphere of radius $r = d\,\bar{N}^{1/3} < d$. This keeps mass bookkeeping exact through both coalescence (merging spheres, $\bar{N} < 1$) and aggregation (sticking, $\bar{N} \geq 1$). The closure is exact for the moments $M_0 = n$ and $M_1 = n\bar{N}$; it approximates the kernel by evaluating it at the mean size.

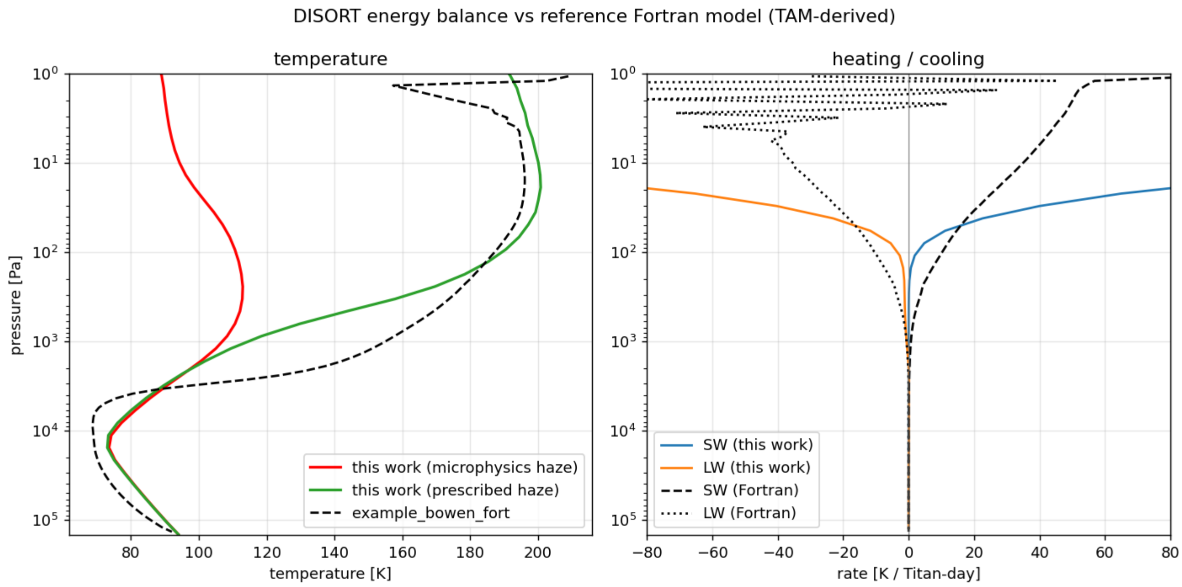

Figure 2: Our DISORT energy balance with the Step 2 *microphysics* haze (red) and with the observational *prescribed* haze (green), versus the reference TAM-lineage correlated-k Fortran model (black), on a shared pressure axis. With the same haze the engines agree (green vs. black); the red–green gap is the microphysics haze. Right: shortwave heating and longwave cooling.

4.2 Fractal geometry and transport coefficients

An aggregate of N monomers of radius d and fractal dimension D has a mass (volume-equivalent) radius and a mobility (drag) radius

$$r(N) = d\,N^{1/3}, \qquad r_a(N) = d\,N^{1/D}, \qquad V(N) = \tfrac{4}{3}\pi d^3 N, \tag{2}$$

with the apparent–bulk conversion $r_a = E^{-1} r^{3/D}$, $E = d^{(D-3)/D}$ (Burgalat and Rannou, 2017). We carry two growth phases explicitly,

$$D(\bar{N}) = \begin{cases} 3 & \bar{N} < 1 \quad \text{(compact coalesced spheres, } E = 1,\ r_a = r\text{)}, \\ D & \bar{N} \geq 1 \quad \text{(fractal aggregates, e.g. } D = 2\text{)}, \end{cases} \tag{3}$$

the switch at $\bar{N} = 1$ being continuous in r_a (both give $r_a = d$).

The Stokes settling velocity with first-order Cunningham slip, generalized to fractal D (Burgalat and Rannou, 2017, Eq. 29), is

$$\omega(N, z) = \frac{2\rho_p\, g(z)\, d^2}{9\,\eta(z)} \left[N^{(D-1)/D} + \frac{A\,\lambda(z)}{d}\, N^{(D-2)/D} \right], \tag{4}$$

with $A = 1.591$, gas viscosity $\eta(z) = \eta(T)$, mean free path $\lambda(z) = \lambda(T, P)$, gravity $g(z)$, and $D = D(\bar{N})$. The equal-size Brownian coagulation kernel is bridged across the Knudsen regimes by the Fuchs harmonic mean of its continuum and free-molecular forms,

$$\beta_{\mathrm{CO}}(N, N) = \frac{2k_B T}{3\eta}\,(r_{ai} + r_{aj}) \left(\frac{1}{r_{ai}} + \frac{1}{r_{aj}} \right) \cdot [\text{slip}], \tag{5}$$

$$\beta_{\mathrm{FM}}(N, N) = \left(\frac{6k_B T}{\rho_p} \right)^{1/2} (r_{ai} + r_{aj})^2 \left(r_i^{-3} + r_j^{-3} \right)^{1/2}, \tag{6}$$

$$\beta(N, N; z) = \frac{\beta_{\mathrm{CO}}\,\beta_{\mathrm{FM}}}{\beta_{\mathrm{CO}} + \beta_{\mathrm{FM}}}. \tag{7}$$

Both ω and β are thus explicit functions of $(\bar{N}, z, T, d, D)$; the temperature enters through $k_B T/\eta$, $(k_B T/\rho_p)^{1/2}$, $\eta(T)$, and $\lambda(T, P)$.

4.3 Governing boundary-value problem

Working with the aggregate number density n and the monomer-volume density $M \equiv n\bar{N}$ (mass $= m_1 M$, $m_1 = \rho_p \frac{4}{3}\pi d^3$), the steady 1-D balance for any density q with downward settling speed ω and eddy diffusivity $K(z)$ reads $\mathrm{d}_z[\omega q + K\,\mathrm{d}_z q] = L_q - S_q$, where $\omega q + K\,\mathrm{d}_z q$ is the downward flux, S_q the production, and L_q the loss. Applied to the two densities,

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[\omega(\bar{N}, z)\, M + K\,\frac{\mathrm{d}M}{\mathrm{d}z}\right] = -\,S_M(z), \tag{8}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[\omega(\bar{N}, z)\, n + K\,\frac{\mathrm{d}n}{\mathrm{d}z}\right] = \tfrac{1}{2}\,\beta(\bar{N}, z)\, n^2 - S_n(z). \tag{9}$$

Equation (8) expresses monomer conservation: coagulation merges particles but preserves total monomer volume, so the only monomer source is production. Equation (9) expresses Smoluchowski number loss—each collision removes one aggregate at rate $\frac{1}{2}\beta n^2$—plus seeding. With $\bar{N} = M/n$ and $D = D(\bar{N})$, this is a coupled two-point boundary-value problem for $M(z), n(z)$. The production source is a Gaussian (de Batz de Trenquelléon et al., 2025, Eq. 5),

$$S_M(z) = \frac{\mathcal{Q}_p}{m_1}\, G(z), \quad G(z) = \frac{1}{\sqrt{2\pi}\,\Delta z}\exp\left[-\frac{(z - z_0)^2}{2\,\Delta z^2}\right], \quad S_n = \frac{S_M}{N_{\mathrm{seed}}}, \tag{10}$$

with $N_{\mathrm{seed}} = (r_p/d)^3$. Boundary conditions are vanishing densities at the model top and settling-only deposition ($\mathrm{d}_z M = \mathrm{d}_z n = 0$) at the surface. A first integral of (8) well below the source recovers the constant downward monomer flux $\omega M + K\,\mathrm{d}_z M = P$, with $P = \mathcal{Q}_p/m_1$.

4.4 Sedimentation-dominated limit: the master scaling ODE

Setting $K \to 0$ yields a closed-form law—both a physical limit and the initial guess for the BVP solver. Monomer-flux constancy gives the classical result that the haze mass density equals the mass flux divided by the fall speed,

$$\omega\, n\, \bar{N} = P \implies n(z) = \frac{P}{\omega\,\bar{N}}, \qquad \rho_h(z) = \frac{\mathcal{Q}_p}{\omega(\bar{N}, z)}. \tag{11}$$

Substituting $\omega n = P/\bar{N}$ into (9) (below the source) collapses the system to a single first-order ODE for the mean size,

$$\boxed{\frac{\mathrm{d}\bar{N}}{\mathrm{d}z} = -\,\frac{\beta(\bar{N}, z)\, P}{2\,\omega(\bar{N}, z)^2}}, \qquad \bar{N}(z_0) = N_{\mathrm{seed}}, \tag{12}$$

integrated downward and switching $D(\bar{N})$ from 3 to D at $\bar{N} = 1$. Because the right-hand side is negative, $\bar{N}$ grows as the particle falls; the number density and sizes then follow from (11) and (2).

4.5 Asymptotic scalings

Holding the background quantities $(T, \eta, \lambda, g, \rho_p, P)$ roughly constant over a scale height, β and ω become pure powers of $\bar{N}$ and (12) integrates to power laws (Table 1). Just below the production region the haze is free-molecular and the mean size grows as $\bar{N} \propto (\mathrm{depth})^2$; deep in the dense lower atmosphere the continuum kernel is size-independent and growth slows to $\bar{N} \propto (\mathrm{depth})^{1/2}$. The full Fuchs kernel (7) interpolates smoothly between these limits, which is why (12)/(8)–(9) are integrated numerically rather than reduced to a single exponent.

Table 1: Asymptotic scaling exponents for equal-size aggregates, with depth $\equiv z_0 - z$. The spherical phase ($\bar{N} < 1$) uses $D = 3$.

Regime	$\beta(\bar{N}) \propto$	$\omega(\bar{N}) \propto$	$\mathrm{d}\bar{N}/\mathrm{d}z \propto -\bar{N}^s$, $s =$	$\bar{N}$(depth), $D = 2$
Continuum (Kn$\ll$1, low alt.)	$\bar{N}^0$	$\bar{N}^{(D-1)/D}$	$-2(D-1)/D$	$\bar{N} \propto \text{depth}^{1/2}$
Free-molecular (Kn$\gg$1, high alt.)	$\bar{N}^{2/D-1/2}$	$\bar{N}^{(D-2)/D}$	$(6-2D)/D - 1/2$	$\bar{N} \propto \text{depth}^2$

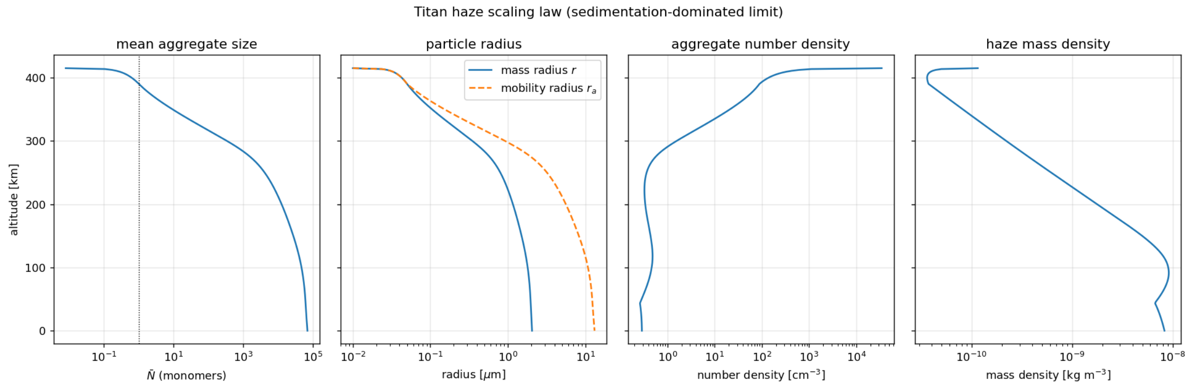

Figure 3: Steady haze profiles from the sedimentation-dominated scaling law (12) on the Titan reference column: mean aggregate size $\bar{N}$ (dotted line marks the $\bar{N} = 1$ spherical/fractal switch), mass and mobility radii, aggregate number density, and haze mass density. Aggregates grow as they settle; r_a exceeds r in the fractal phase.

4.6 Numerical solution

We integrate the master ODE (12) downward from $z_0 = 415\,\text{km}$ (seed $N_\text{seed} = (r_p/d)^3 \approx 8 \times 10^{-3}$) on the crude Titan reference column, with $\eta(T)$ from a Sutherland law and $\lambda(T, P)$ from kinetic theory (Table 2). Figure 3 shows the result. The mean size grows by seven orders of magnitude from the sub-monomer seed to $\bar{N} \sim 7 \times 10^4$ at the surface; the spherical$\to$fractal switch at $\bar{N} = 1$ near 390 km is visible as the divergence of the mobility radius r_a from the mass radius r (open aggregates have $r_a > r$ for $D < 3$). The mass radius reaches $\approx 0.35\,\mu\text{m}$ near 300 km—consistent with the characteristic radius $r_c \approx 0.46\,\mu\text{m}$ of de Batz de Trenquelléon et al. (2025)—and $\approx 2\,\mu\text{m}$ at the surface. Number densities span $\sim$0.3–$3 \times 10^4\,\text{cm}^{-3}$, and the monomer mass flux $\rho_h\,\omega$ is conserved to machine precision, confirming Eq. (11).

4.7 Eddy-diffusion BVP and cross-validation

Restoring eddy diffusion, we solve the full boundary-value problem (8)–(9) on $[0, z_0]$, imposing the (narrow) production as a top flux boundary condition—downward monomer flux $\Phi_M = P$ and seed number flux $\Phi_n = P/N_\text{seed}$—and settling-only deposition at the surface. The $K \to 0$ master ODE supplies the initial guess. A Titan-like eddy profile $K(z) = K_0 \exp(z/H_K)$ with $K_0 = 1\,\text{m}^2\text{s}^{-1}$ and $H_K = 150\,\text{km}$ keeps K small ($\lesssim 16\,\text{m}^2\text{s}^{-1}$) across the haze, as expected in Titan's stratosphere.

We cross-validate the resulting profile against two published constraints (Fig. 4). First, the haze extinction scale height: fitting $n\,r_a^2$ over 80–300 km yields $H = 64\,\text{km}$, against the 65 km inferred by Tomasko et al. (2008); the $K \to 0$ settling limit alone gives 54 km, so the modest eddy diffusion brings the model into close agreement. Second, the characteristic size: the mass radius is $\approx 0.34\,\mu\text{m}$ near 300 km and sub-micron through the main haze, consistent with the $r_c \approx 0.46\,\mu\text{m}$ of de Batz de Trenquelléon et al. (2025). The monomer mass flux is conserved to machine precision (global mass balance: surface deposition equals column production), and

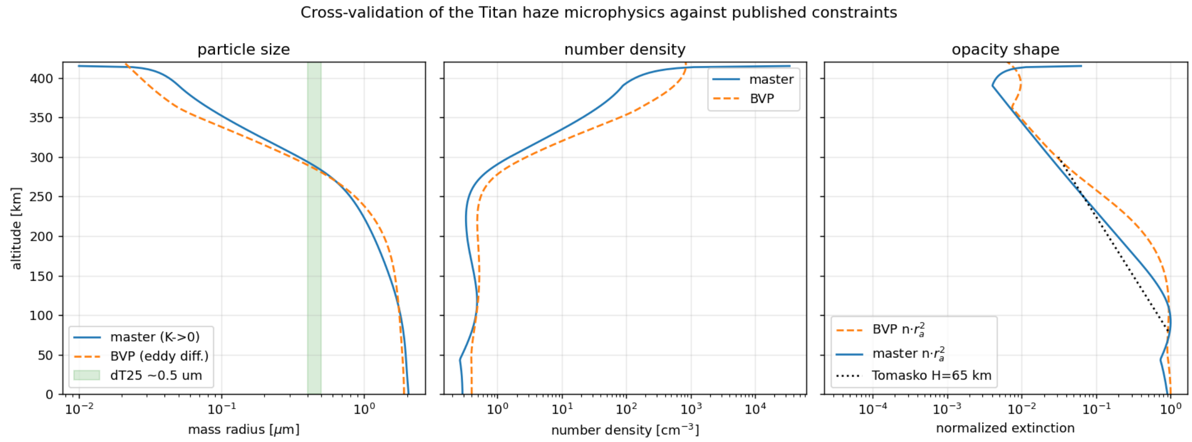

Figure 4: Cross-validation of the microphysics against published Titan constraints. Left: mass radius from the $K \to 0$ master ODE and the eddy-diffusion BVP, with the de Trenquelléon et al. (2025) main-haze size ($\sim 0.5\,\mu$m) shaded. Centre: number density. Right: normalized extinction $n\,r_a^2$, with the Tomasko et al. (2008) $H = 65$ km reference slope (dotted) anchored at 80 km.

the BVP departs from the master ODE by only $\sim 9\%$ in the settling-dominated lower haze. The extinction scale height is sensitive to $K(z)$, identifying the eddy profile as the principal uncertain input.

5 Coupled iteration (Step 3)

Per layer the microphysics supplies the number density $n(z)$ and the characteristic radius $r_a(z) = d\,\bar{N}^{1/D}$. These map to layer optical properties via a precomputed aggregate-optics table (de Batz de Trenquelléon et al., 2025, Eqs. 16–18): the optical thickness is $\Delta\tau(\lambda, z) = n(z)\,C_{\text{ext}}(r_a, \lambda)\,\Delta z$, with mean single-scattering albedo and asymmetry parameter obtained by opacity weighting. The aggregate projected area scales as $\bar{N}^{2/D} d^2$. The feedback loop is then

$$T(z) \xrightarrow{\text{DISORT}} \eta(T), \lambda(T, P) \xrightarrow{(12)/\S 4} n(z), \bar{N}(z) \xrightarrow{\text{optics}} \Delta\tau, \omega_0, g \xrightarrow{\text{DISORT}} T(z),$$

iterated to a fixed point. The sign and magnitude of the feedback gain (e.g. $\mathrm{d}T/\mathrm{d}\tau$ at convergence) are read off directly from the converged sequence.

6 Photochemistry in the loop (Step 4)

In Steps 1–3 the production rate $\mathcal{Q}_p$ and seed properties are imposed ($\mathcal{Q}_p = 2.1 \times 10^{-13}\,\mathrm{kg\,m^{-2}\,s^{-1}}$ at $z_0 \approx 415$ km; de Batz de Trenquelléon et al., 2025). Step 4 promotes $\mathcal{Q}_p$ to an output of a photochemical module driven by CH_4/N_2 photolysis, so the haze mass flux responds to the radiation field and composition, completing the radiation↔chemistry↔microphysics feedback.

7 Baseline parameters

Status and approximations. The monodisperse closure replaces the true size spread by $\bar{N}$ and evaluates the kernel at the mean size; real distributions coagulate somewhat faster, an $O(1)$ correction absorbable into a prefactor on β and calibratable against a full population-balance run. The spherical→fractal switch at $\bar{N} = 1$ is treated as instantaneous. Background profiles $\eta(T)$, $\lambda(T, P)$ are taken from N_2 kinetic theory, $g(z)$ from Titan gravity, and the eddy diffusivity

Table 2: Baseline microphysical and atmospheric parameters adopted for Titan. Sources: T08 (Tomasko et al., 2008), L23 (Lombardo and Lora, 2023), BR17 (Burgalat and Rannou, 2017), dT25 (de Batz de Trenquelléon et al., 2025).

Quantity	Symbol	Value	Source
Fractal dimension	D	2.0 (free, 2–3; 3 $\Rightarrow$ spheres)	dT25
Monomer radius	d	50 nm	dT25 (T08)
Production seed radius	r_p	10 nm	dT25
Aerosol density	ρ_p	$800\,\mathrm{kg\,m^{-3}}$	dT25
Mass production rate	$\mathcal{Q}_p$	$2.1 \times 10^{-13}\,\mathrm{kg\,m^{-2}\,s^{-1}}$	dT25 (T08)
Production altitude	z_0	$415\,\mathrm{km} \approx 1\,\mathrm{Pa}$	dT25
Production width	Δz	20 km	dT25
Charge density	n_e	$15\,e^-\,\mu\mathrm{m}^{-1}$	dT25
Slip-flow constant	A	1.591	BR17
Surface temperature	T_s	93.5 K	T08
Surface pressure	P_s	$\sim$1.47 bar	L23
Stratospheric CH_4	–	1.4%	T08
Haze opacity scale height	H	65 km (above 80 km)	T08
RT grid	–	0–540 km, $\sim$50 layers	T08, L23
Shortwave / longwave split	–	$5\,\mu$m	L23

$K(z)$ from a standard Titan profile (Vuitton et al., 2019). Charge inhibition (a multiplicative factor on β) is deferred until the base loop closes. Tables 1 and 2 and Eqs. (8)–(12) define the model to be implemented next.

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